

1. Let $R = \text{Fun}(\mathbb{R}, \mathbb{R})$ be the ring in exercise D2 of the "Ring Basics" adventure sheet. 0_R and 1_R are the constant functions zero and one.

Show which of the following subsets of R are subrings of R . If they are not subrings, show whether they are rings (with a different multiplicative identity than 1_R , but endowed with the same operations as in R) or not.

- (a) The set C of constant functions.
- (b) The set S of those functions f such that $f(q) = 0$ for any $q \in \mathbb{Q}$.
- (c) The set T consisting of 0_R , together with those functions with no zeros, or only a finite number of zeros. (A zero of a function $f \in R$ is an element $x \in \mathbb{R}$ such that $f(x) = 0$)

(a) C is a subring of R .

Pf. Since 0_R is $f(x) = 0$, 1_R is $f(x) = 1$,

$$\underline{0_R, 1_R \in C} \quad \textcircled{1}$$

Let f, g be two element in C ,

Suppose $f(x) = a$, $g(x) = b$ for some $a, b \in \mathbb{R}$

$$\text{Then } (f+g)(x) = a+b = f(x)+g(x)$$

so C is closed under addition $\textcircled{2}$

$$fg(x) = ab = f(x) \cdot g(x)$$

so C is closed under multiplication $\textcircled{3}$

Since $-f(x) = -a$ is also an constant function,

$-f \in C$, so C is closed under additive inverse $\textcircled{4}$.

Since $C \subseteq R$ and R is a ring, by worksheet,
it suffices to show $\textcircled{1} \textcircled{2} \textcircled{3} \textcircled{4}$

So C is a subring of R .

(b) S is not a subring of R .

Since 1_R which is the constant function $f(x) = 1$ is not in S (for $x \in \mathbb{Q}$, $f(x) \neq 0$, so f is not in S)

S violates the definition of subring.

And S is not even a ring because it does not have a multiplicative identity.

To show this, we assume there is a function

$f \in S$ such that $\forall g \in S$, $fg = gf = g$

Take $g(x) = 2$

So for any $x \in \mathbb{R}$, $2f(x) = 2 \Rightarrow f(x) = 1$,
which is not in S .

So S does not have a multiplicative inverse,
therefore is not a ring.

(c) T is not a subring of R , and not a ring.

Consider $f(x) = \begin{cases} 1, & x \geq 0 \\ x, & x < 0 \end{cases}$

$g(x) = \begin{cases} -1, & x \geq 0 \\ x, & x < 0 \end{cases}$

So $f(x)$ and $g(x)$ both only contains
one zero point, therefore $\underline{f(x), g(x) \in T}$

But $f(x) + g(x) = \begin{cases} 0, & x \geq 0 \\ 2x, & x < 0 \end{cases}$ which
contains infinitely many "zero's".

So $\underline{f(x) + g(x) \notin T}$

So T is not closed with addition, so

T is not a ring and definitely not a
subring of R .

2. An element x in a ring R is said to be *nilpotent* if $x^m = 0_R$ for some positive integer m . Generalizing the definition on page 40 of our text, a *unit* u in a ring R is an element with a multiplicative inverse, meaning there exists $s \in R$ such that $su = us = 1_R$.

- (a) Prove that if $x \in R$ is nilpotent (and R is not the zero ring), then x cannot be a unit.
- (b) Prove that if $x \in R$ is nilpotent, then $(1_R - x)$ is a unit. (Hint: One approach to showing something is a unit is to write down its inverse. In this case, it could help to recall geometric series from Calculus.)
- (c) Describe all the nilpotent elements in $\mathbb{Z}_n$ in terms of their prime factorization.

(a) Let R be a ring which is not the zero ring
($0_R, 1_R$ are different elements).

Assume $x \in R$ is a nilpotent.

Then for some $m \in \mathbb{Z}$, $x^m = 0_R$.

Let m be the smallest positive integer such that
Case 1: $m \geq 2$ $x^m = 0_R$.

Assume for sake of contradiction that x is a unit. Then for some $y \in R$, $xy = yx = 1_R$.

Then we multiply both sides of $xy = yx$ by x^{m-1}

$$x^{m-1} \cdot xy = x^{m-1} \cdot yx$$

$$\Rightarrow (x^m)y = x^{m-1}(yx) \text{ by associativity of multiplication in } R$$

$$\Rightarrow 0_R \cdot y = x^{m-1} \cdot 1_R$$

$$\Rightarrow 0_R = x^{m-1} \text{ by definition of } 1_R, 0_R$$

This violates the assumption that m is the smallest integer s.t. $x^m = 0_R$.

Case 2 : $m = 1$

Then $x = 0_R$, so x can not be a unit (since $0_R \neq 1_R$, $\forall x \in R$, $xy = 0_R y = 0_R \neq 1_R$)

This contradicts with that x is a unit.

Since every case causes contradiction, we have proved that if $x \in R$ is a nilpotent, then x is not a unit.

(b) Let $x \in R$ be a nilpotent.

Let m be the smallest positive integer s.t. $x^m = 0_R$
By definition,

Case 1 : $m=1$, then $x = 0_R$

Consider 1_R , then $1_R(1_R - x) = 1_R \cdot 1_R = (1_R - x) \cdot 1_R$

So $(1_R - x)$ is a unit. $= 1_R$

Case 2 : $m \geq 2$

Consider $y = 1_R + x + x^2 + \dots + x^{m-1}$

$$\begin{aligned} \text{Then } (1_R - x)y &= 1_R + x + x^2 + \dots + x^{m-1} \\ &\quad - x - x^2 - \dots - x^{m-1} - x^m \\ &= 1_R - x^m = 1_R - 0_R = 1_R \end{aligned}$$

Similarly we get $y(1_R - x) = 1_R$

So $(1_R - x)$ is a unit.

Therefore we have proved the statement.

(3) By FTA, $n = \underbrace{p_1^{a_1} \cdot p_2^{a_2} \cdot p_3^{a_3} \cdots p_k^{a_k}}_{p_1 \cdots p_k \text{ and their multiplicity } a_1 \cdots a_k}$ for some primes

So for any nilpotent x of $\mathbb{Z}_n$.

$$x^\alpha \equiv 0 \pmod{n} \text{ for some } \alpha \in \mathbb{Z}$$

$$\text{So } x^\alpha = \beta n = \beta p_1^{a_1} \cdot p_2^{a_2} \cdots p_k^{a_k} \text{ for some } \beta \in \mathbb{Z}.$$

Therefore $(p_1^{a_1} \cdot p_2^{a_2} \cdots p_k^{a_k}) \mid x^\alpha$.

So let x contain all prime factors $p_1 \cdots p_k$

And consider $\alpha = \max(a_1, a_2, \dots, a_n)$,

it ensures that x^α contains $(p_1^{a_1} \dots p_k^{a_k})$ as

Therefore we have proved that as long as x contains all prime factor of n , x is different a nilpotent

Note that

x also must contain all prime factors, that

is, if some prime $p_i | n$ but $p_i \nmid x$, then

x is not a nilpotent. This is obvious since if $p_i \nmid x$, there is no $d \in \mathbb{Z}$ s.t. x^d has the factor $p_i^{a_i}$ of n .

So the set of all nilpotents of $\mathbb{Z}_n$ is just the set of multiples of all different prime factors of n

$$\left\{ [x]_n \mid x = t(p_1 \dots p_k), t \in \mathbb{Z}, p_1 \dots p_k \text{ are all different prime factors of } n \right\}$$

3. An element $r \neq 0$ in a commutative ring R is said to be a *zerodivisor* if there exists a nonzero element $s \in R$ such that $rs = 0$.

- (a) Given a nonzero element $r \in R$, prove that r is not a zerodivisor if and only if the map $R \rightarrow R$ given by multiplication by r , meaning the map $s \mapsto rs$, is injective.
- (b) Describe all the zerodivisors in $\mathbb{Z}_n$ in terms of the prime factorization of n or their greatest common divisor with n .

(A) Denote the map by $f(s) = rs$.

① Assume r is not a zerodivisor.

Then assume $f(s_1) = f(s_2) \Rightarrow rs_1 = rs_2$.

$$\Rightarrow r(s_1 - s_2) = 0_R \quad (\text{and } r \neq 0_R)$$

Since r is not a zerodivisor,

there is no nonzero element s such that $rs = 0_R$.

So $(s_1 - s_2)$ can only be 0_R .

So $s_1 = s_2$

Hence $f(s_1) = f(s_2) \Rightarrow s_1 = s_2$, the function is injective.

② Assume f is injective

Assume for contradiction that r is a zerodivisor.

Then for some $s \in R$ with $s \neq 0_R$, $sr = 0_R$.

$$\text{So } f(s) = 0_R$$

$$\text{And since } f(0_R) = r \cdot 0_R = 0_R$$

$f(s) = f(0_R)$ while $s \neq 0_R$, contradicting with

Hence r is not a zero division f is injective

Since ①, ②, we have proved the iff statement.

(b) Let $[r]_n$ be a zero divisor in $\mathbb{Z}_n$
 It means $\exists [s]_n \in \mathbb{Z}_n$ s.t. $[r]_n [s]_n = [0]_n$
 which is not $[0]_n$

$$\Leftrightarrow rs = kn \text{ for some } s, k \in \mathbb{Z} \text{ with } n \nmid s.$$

(1) if $\gcd(r, n) = 1$, then r, n has no common prime factor

this means that to satisfy $rs = kn$,

s must contain all prime factors of n ,

this means $n \mid s$.

So the circumstance is impossible.

(2) If $\gcd(r, n) > 1$, then r, n has at least some common factor p .

By FTA, $n = p \cdot (q_1 \dots q_\alpha)$ for some primes $q_1, \dots, q_\alpha$

So consider taking $s = q_1 \dots q_\alpha$, then rs contain

all prime factors of n , which means $rs = kn$ for some $k \in \mathbb{Z}$,

then $[s]_n$ is a solution to $[r]_n X = [0]_n$.

And here $s = q_1 \dots q_\alpha < n$ so $n \nmid s$, satisfying the requirement that $[s]_n \neq [0]_n$

Therefore, the set of all zero divisors of $\mathbb{Z}_n$ is

just $\boxed{\{[r]_n \mid \gcd(r, n) > 1\}}$.

4. For two rings, R and S a function $\varphi: R \rightarrow S$ is a *ring homomorphism* if $\varphi(1_R) = 1_S$, and for all $x, y \in R$ the following conditions hold

$$\varphi(x +_R y) = \varphi(x) +_S \varphi(y), \text{ and } \varphi(x \times_R y) = \varphi(x) \times_S \varphi(y)$$

- (a) Let R be any ring (recalling how our class convention differs from that of the book!). Prove that there exists a unique ring homomorphism $\mathbb{Z} \rightarrow R$.
- (b) Let $n > 1$ be an integer. Prove that there does not exist a ring homomorphism $\mathbb{Z}_n \rightarrow \mathbb{Z}$.
- (c) Suppose R and S are two rings, and $f: R \rightarrow S$ is a ring isomorphism; in particular, f is a bijection and so has an inverse function $g: S \rightarrow R$. Prove that g is also a ring homomorphism.
- (d) Prove the following theorem that appears in our worksheets:

If $f: R \rightarrow S$ is a ring homomorphism, then f is injective if and only if $\ker f = \{0_R\}$.

(a) Consider $\varphi: \mathbb{Z} \rightarrow R$
 $n \mapsto n \cdot 1_R$

$$\text{So } \varphi(1_{\mathbb{Z}}) = 1_R$$

Let x, y be two arbitrary elements in $\mathbb{Z}$

$$\begin{aligned} \text{Then } \varphi(x+y) &= (x+y) \cdot 1_R = x \cdot 1_R + y \cdot 1_R = \varphi(x) + \varphi(y) \\ \varphi(xy) &= (xy) \cdot 1_R = (x \cdot 1_R) \cdot (y \cdot 1_R) = \varphi(x) \cdot \varphi(y) \end{aligned}$$

So φ is a homomorphism

Assume f is any homomorphism from $\mathbb{Z}$ to R

$$\text{Then } f(1) = \varphi(1) = 1_R$$

And by theorem 3.10 on textbook,

$$f(-1) = -f(1) = -\varphi(1) = -1_R$$

$$\text{and } f(0) = \varphi(0) = 0_R.$$

Let n be an arbitrary positive integer that is not 1.

By the definition of homomorphism,

$$f(n) = f(\underbrace{1+1+\dots+1}_{n \text{ times}}) = \underbrace{f(1)+f(1)+\dots+f(1)}_{n \text{ times}}$$

$$\text{So } f(n) = n f(1) = n \cdot 1_{\mathbb{R}} = \varphi(n)$$

Similarly, for any negative integer m that is not -1 ,

$$\begin{aligned} f(m) &= f(\underbrace{(-1)+(-1)+\dots+(-1)}_{-m \text{ times}}) = \underbrace{f(-1)+f(-1)+\dots+f(-1)}_{-m \text{ times}} \\ &= -m f(-1) \\ &= m \cdot 1_{\mathbb{R}} = \varphi(m) \end{aligned}$$

Therefore for any $n \in \mathbb{Z}$, $\varphi(n) = f(n)$

So $\varphi = f$.

Therefore the homomorphism is unique.

(b) Assume for sake of contradiction that φ is a homomorphism from $\mathbb{Z}_n$ to $\mathbb{Z}$.

By definition of homomorphism, $\varphi([0]) = 0$,
 $\varphi([1]) = 1$.

$$\text{So } \underline{\varphi([1]_n + [1]_n) = \varphi([1]_n) + \varphi([1]_n) = 2} \quad \textcircled{1}$$

Suppose $1_2 = [a]_n \in \mathbb{Z}_n$

Repeat process ① by n times.

Then we get $\varphi(\underbrace{[1]_n + [1]_n + \dots + [1]_n}_{n \text{ times}}) = n$

$$\text{So } \varphi([n]_n) = n$$

Since $[n]_n = [0]_n$, $\varphi([0]_n) = n$ which contradicts with $\varphi([0]_n) = 0$, violating definition of homomorphism φ as a function

Therefore such homomorphism does not exist.

$$(c) f: R \rightarrow S$$

Since $f: R \rightarrow S$ is a ring isomorphism,

let s_1, s_2 be two arbitrary elements in S .

Since f is bijective, there exists unique element

$$r_1, r_2 \in R \text{ s.t. } f(r_1) = s_1, f(r_2) = s_2$$

$$\begin{aligned} \text{So } \underbrace{f^{-1}(s_1 + s_2)}_{g(s_1 + s_2)} &= f^{-1}(f(r_1) + f(r_2)) \stackrel{(\text{since } f \text{ is homomorphism})}{=} \underbrace{f^{-1}(f(r_1 + r_2))}_{f^{-1}(s_1 + s_2)} \\ &= r_1 + r_2 = f^{-1}(s_1) + f^{-1}(s_2) \\ &= g(s_1) + g(s_2) \end{aligned}$$

So g is closed under addition, ①

$$\begin{aligned}
 g(s_1, s_2) &= f^{-1}(s_1 s_2) = f^{-1}(f(r_1) \cdot f(r_2)) \\
 &= f^{-1}(f(r_1 r_2)) \text{ since } f \text{ is homomorphism} \\
 &= r_1 r_2 \text{ by property of inverse function.} \\
 &= f^{-1}(s_1) \cdot f^{-1}(s_2) \\
 &= g(s_1) g(s_2)
 \end{aligned}$$

So g is closed under multiplication ②

Also, since f is homomorphism, $f(1_R) = 1_S$

Also since f is bijection and has inverse,

$$\underline{1_R = f^{-1}(1_S) = g(1_S)} \quad \text{③}$$

By ①, ②, ③, g is also a ring homomorphism.

(d) First we prove: if f is injective then $\ker(f) = \{0_R\}$

Assume (Since $f(0_R) = 0_S$ by f being a homomorphism,
 f is injective $0 \in \ker(f)$.)

Let $r \in \ker(f)$, so $f(r) = 0_S = f(0_R)$

Since f is injective, $f(r) = f(0_R)$ implies

So any element in $\ker(f)$ can only ^{$r = 0_R$} be 0_R

So $\ker(f) = 0_R$. $\square$

Next we prove: if $\ker(f) = \{0_R\}$ then f is injective.

Assume $\ker(f) = \{0\}$

Let $s_1 = f(r_1)$, $s_2 = f(r_2)$ and $s_1 = s_2$
($f(r_1) = f(r_2)$)

$$\text{So } f(r_1) - f(r_2) = 0_S$$

Since f is a homomorphism,

$$f(r_1) - f(r_2) = f(r_1 - r_2) = 0_S$$

$$\text{So } (r_1 - r_2) \in \ker(f)$$

$$\text{Since } \ker(f) = \{0_R\}, \quad r_1 - r_2 = 0_R \Rightarrow r_1 = r_2$$

□

Therefore, f is injective iff $\ker(f) = \{0_R\}$